

A Project report on

HEALTH COVERAGE ADVISOR

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the academic requirements for the award of the degree.

Bachelor of Technology
in
Computer Science and Engineering

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CERTIFICATE

This is to certify that the Mini Project II report entitled "**HEALTH COVERAGE ADVISOR**" being submitted by D. Sai Sujan (21H51A0534), D.Sahithi (21H51A0535), D. V. S. S. Jaya Prakash (21H51A0536) in partial fulfilment for the award of **Bachelor of Technology in Computer Science and Engineering** is a record of Bonafide work carried out his/her under my guidance and supervision.

The results embodied in this project report have not been submitted to any other University or Institute for the award of any Degree.

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ABSTRACT

In the realm of healthcare insurance, choosing the right coverage can be overwhelming due to the multitude of options and individual-specific considerations. The "Health Coverage Advisor" project addresses this challenge by employing a sophisticated blend of machine learning algorithms and web scraping techniques. By integrating user provided health information, medical history, and Body Mass Index (BMI). The system autonomously evaluates and compares diverse insurance policies available from various providers. Through this data-driven approach, the project aims to streamline the insurance selection process, providing users with personalized recommendations that align closely with their healthcare needs and financial preferences. The core functionality of the "Health Coverage Advisor" revolves around its ability to harness large-scale data scraping from insurance providers' websites. This process aggregates up-to-date policy details, ensuring that the recommendations offered to users are based on the latest information available in the market. Machine learning models then analyse these datasets alongside user profiles, identifying patterns and correlations that optimize policy matches. This not only saves users significant time and effort in manually researching insurance options but also enhances decision-making by presenting tailored suggestions that account for individual health risks, budget constraints, and coverage preferences.

CHAPTER 1

INTRODUCTION

CHAPTER 1

INTRODUCTION

1.1. Problem Statement

Selecting the most appropriate healthcare insurance plan is a complex and time-consuming process due to the vast number of available policies, frequent updates to coverage details, and the individualized nature of healthcare needs. Users are often overwhelmed by the need to manually compare policies, which may result in suboptimal choices that fail to meet their specific health requirements or financial preferences.

Traditional comparison tools offer limited personalization, as they do not take into account critical user-specific factors like medical history, health risks, and budget constraints. Moreover, the dynamic nature of the insurance market, where policy terms frequently change, adds an additional layer of complexity.

To address this challenge, there is a need for an automated system that can evaluate and compare insurance policies in real-time, considering individual health data, medical history, and BMI. Such a system would streamline the insurance selection process by providing personalized recommendations based on the most current policy details, helping users make informed decisions that are tailored to their unique healthcare needs.

1.2. Research Objective

- Develop a web scraping pipeline that continuously collects and updates healthcare insurance policy data from multiple providers, ensuring the information is current and comprehensive.
- Implement machine learning models that accurately analyse user-specific health profiles, including factors such as medical history and BMI, to identify and recommend suitable insurance policies.
- Integrate real-time updates into the system to ensure that policy recommendations are always based on the latest market data and changes in insurance terms.
- Assess the system's performance in terms of accuracy, efficiency, and user satisfaction by testing its ability to provide personalized, relevant insurance recommendations tailored to individual healthcare needs and financial preferences.

1.3. Project Scope and Limitations

Scope of the Project:

- **Data Collection:** Develop a web scraping system to continuously collect and update healthcare insurance policy data from various providers.
- **User Health Profiling:** Gather and analyse user health profiles, including medical history, BMI, and financial preferences, to tailor insurance recommendations.
- **Machine Learning Integration:** Implement machine learning algorithms to match users with the most suitable insurance plans based on their health data and current market offerings.
- **Personalized Recommendations:** Provide personalized, real-time insurance recommendations that adjust to changing policy details and user input.
- **System Evaluation and Optimization:** Evaluate and optimize system performance, ensuring scalability and accuracy in delivering relevant and up-to-date insurance options for a wide range of users.

Limitations:

- The system's effectiveness depends on the accuracy and completeness of the scraped insurance data, which may lead to suboptimal recommendations.
- Legal restrictions on web scraping and compliance with data protection regulations (are essential when handling sensitive user health information.
- Machine learning models may struggle to generalize for users with unique health conditions, and rapid changes in insurance policies could impact timely, relevant recommendations.

CHAPTER 2

BACKGROUND

WORK

CHAPTER 2

BACKGROUND WORK

2.1. Basic Coverage Calculators

2.1.1. Introduction

Basic coverage calculators are online tools that estimate health insurance coverage needs using user inputs like age, health status, and lifestyle factors.

2.1.2. Merits, Demerits, and Challenges

Merits:

They are easily accessible and can quickly give users a ballpark figure of the coverage they might need without requiring extensive information.

Demerits:

These calculators often lack detailed personalization and may not take into account unique health conditions or specific insurance products that best fit the user.

Challenges:

The accuracy of the estimates can vary widely, leading to potential gaps in coverage that user may not anticipate.

2.1.3. Implementation of Basic Coverage Calculators

These tools involve creating a user-friendly interface and developing algorithms that can process user inputs to generate estimates.

2.2. Financial Planning Services

2.2.1. Introduction

Financial planning services offer professional advice on estimating healthcare coverage needs as part of a comprehensive financial strategy.

2.2.2. Merits, Demerits, and Challenges

Merits:

These services provide personalized insights and consider users' overall financial situations, helping clients make informed decisions about their insurance needs.

Demerits:

They often lack real-time data and automated analysis, resulting in slower response times and potentially outdated information.

Challenges:

Users might face high costs for these services, which can be a barrier for those seeking affordable advice.

2.2.3. Implementation of Financial Planning Services

Financial planning services involves recruiting qualified advisors and developing a structured process for evaluating client needs and providing recommendations.

2.3. Health Risk Assessments**2.3.1. Introduction**

Health risk assessment platforms evaluate users' health risks and provide general guidance on appropriate insurance coverage.

2.3.2. Merits, Demerits, and Challenges**Merits:**

They can help users understand their health status and potential risks, leading to more informed decisions about coverage needs.

Demerits:

These platforms may not provide detailed cost projections or personalized limiting their overall effectiveness.

Challenges:

The assessments might lack comprehensive data integration, leading to generic advice that does not consider individual circumstances.

2.3.3. Implementation of Health Risk Assessments

These platforms require collecting relevant health data and developing algorithms to analyze risks and suggest general coverage guidance. The platform presents the results via an intuitive interface, offering personalized health insights and preventive recommendations. It also integrates real-time updates from healthcare sources to ensure the accuracy and relevance of the assessments, enabling users to make informed decisions about their health.

CHAPTER 3

PROPOSED SYSTEM

CHAPTER 3

PROPOSED SYSTEM

3.1. Objective of Proposed Model:

Health Coverage Advisor aims to empower individuals to make informed healthcare insurance decisions by providing personalized recommendations based on user-specific health data, medical history, and Body Mass Index (BMI). Utilizing advanced machine learning algorithms and real-time data scraping from various insurance providers, the model seeks to aggregate the latest policy details and streamline the evaluation process. This approach not only saves users significant time and effort in comparing options but also enhances decision-making by matching them with policies that align with their healthcare needs and financial preferences, ultimately promoting better health outcomes.

3.2. Algorithm used for Proposed Model:

Health Coverage Advisor model employs a combination of machine learning algorithms and data processing techniques to deliver personalized insurance recommendations. Initially, data preprocessing algorithms such as normalization and missing value imputation are utilized to ensure that user health metrics and policy features are on a comparable scale. For classification, decision trees and random forest algorithms are employed to classify users based on their health data and match them with suitable insurance policies, providing robust and interpretable recommendations. To further enhance personalization, clustering algorithms like K-means segment users into distinct groups based on similarities in health profiles. The model also integrates collaborative and content-based filtering techniques to recommend policies that align with user preferences and behaviors. Additionally, natural language processing is utilized for sentiment analysis of user feedback on insurance policies, refining recommendations based on customer satisfaction levels. Finally, performance is evaluated using metrics like accuracy, precision, recall, and F1-score, ensuring that the model delivers reliable and effective policy suggestions tailored to each user's unique healthcare needs and financial situations.

3.3 Designing:

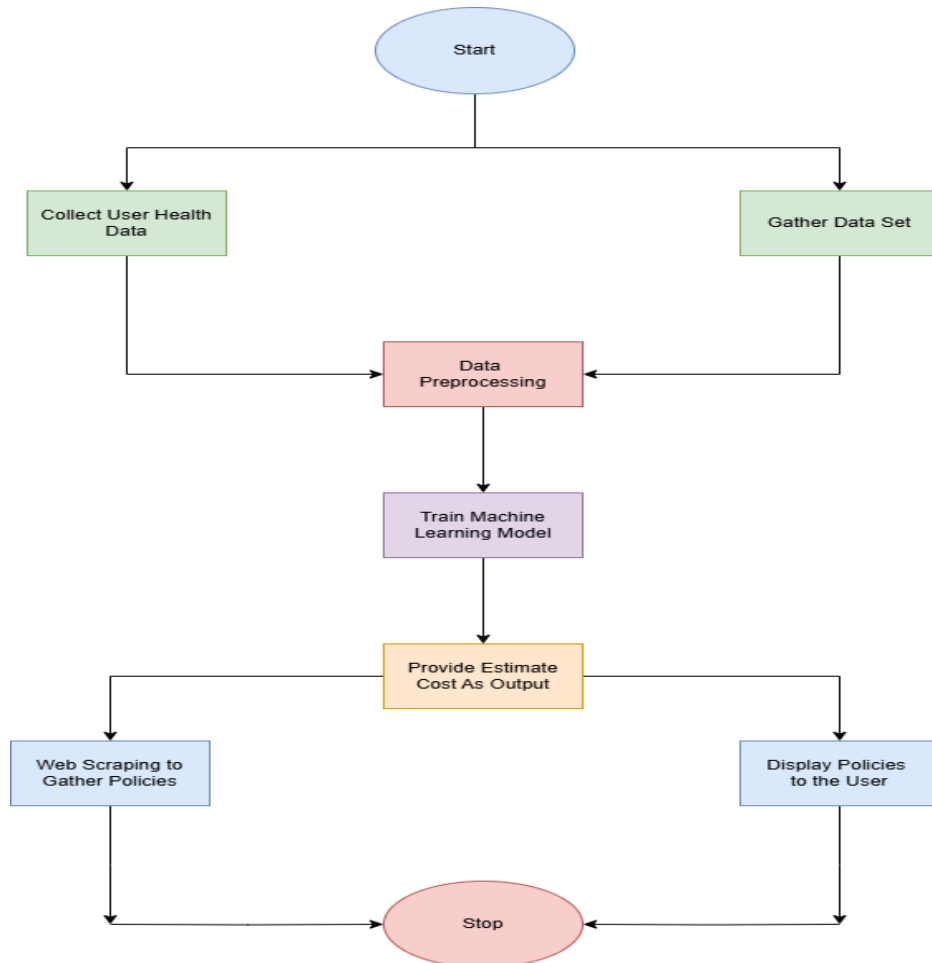


Fig.3.1. System Architecture

3.3.1 Flow Chart Diagram:

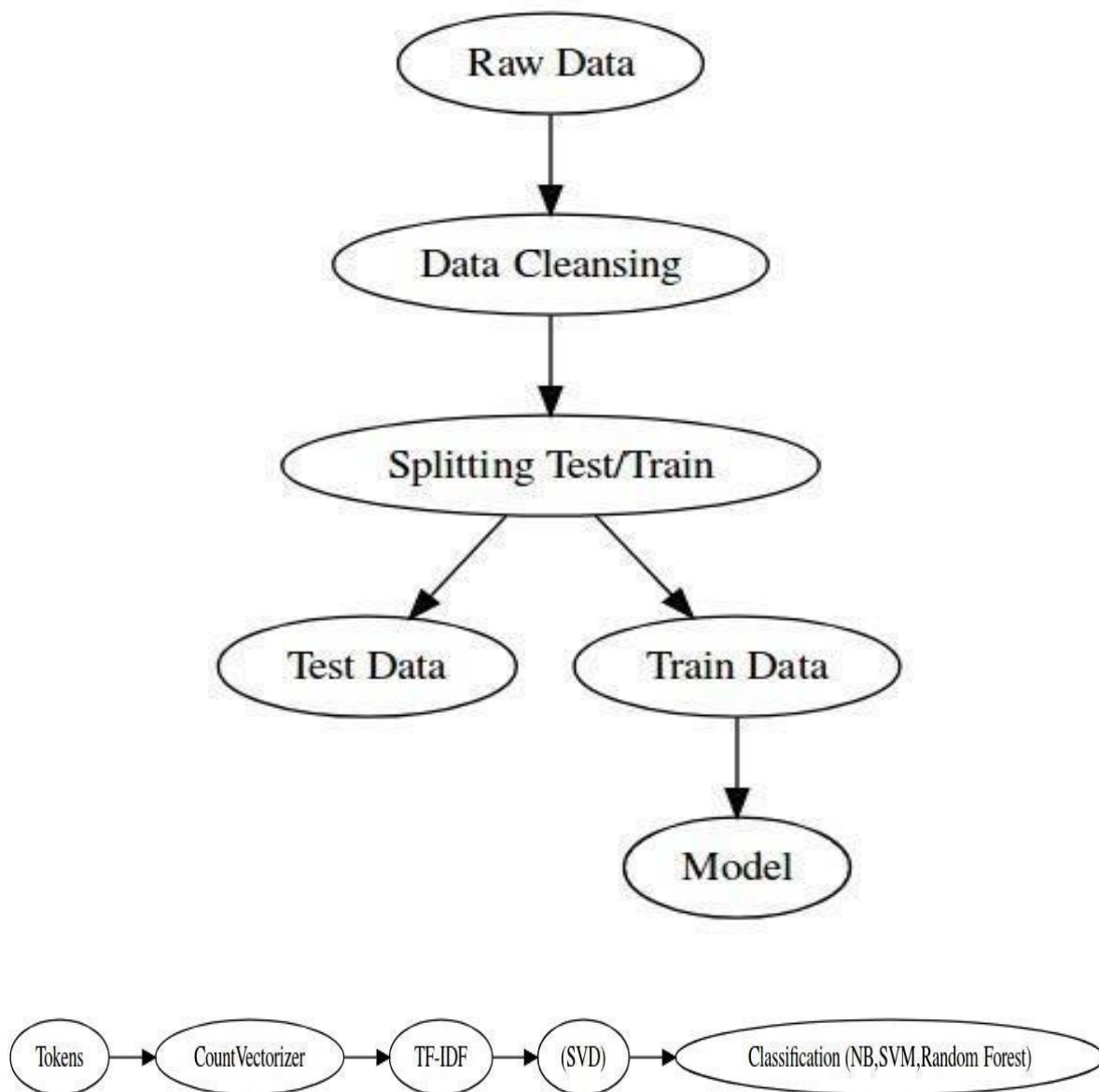


Fig 3.2 Flow chart Diagram

3.4 Step Wise Implementation and Code:

3.4.1. Data Collection:

Scrape data from various insurance providers' websites to gather real-time information on available policies, coverage details, and pricing. Collect user health data through secure forms, ensuring privacy and compliance with regulations.

3.4.2. Data Cleaning:

Remove duplicates and irrelevant entries from the collected datasets. Address inconsistencies in data formatting (e.g., date formats, categorical values). Handle missing values using appropriate imputation techniques (e.g., mean/mode substitution or k-NN) or remove records with excessive missing information.

3.4.3. Exploratory Data Analysis (EDA):

Perform EDA to understand the data distribution, identify patterns, and detect anomalies. Analyse correlations between user health metrics and insurance policy attributes to gain insights into significant relationships that could influence recommendations. Plotted histograms for continuous variables such as age, BMI, and claim. Created count plots for categorical variables like gender, diabetic, smoker, and region. Computed the correlation matrix. Visualized the correlation matrix using a heatmap.

3.4.4. Data Preprocessing:

Normalize and standardize numerical data (e.g., BMI) to ensure comparability. Encode categorical variables using techniques like one-hot encoding.

3.4.5. Model Building:

Build a machine learning model using Random Forest algorithm.

Train the model on the preprocessed data and evaluate its performance using metrics such as accuracy, precision, recall, and F1-score.

3.4.6. Website Creation:

Create a website using Streamlit, which allows you to create interactive web applications with Python.

CODE

```
import streamlit as st
import pandas as pd
import pickle
import numpy as np

st.markdown(
    """
    <div>
        <center><h1>Medical Cost Predictor</h1></center>
    </div>
    """,
    unsafe_allow_html=True,
)
st.write("### Information of the beneficiary")
model = pickle.load(open("Model/model.pkl", "rb"))

input_data = []
# Age
input_data.append(st.slider("Enter his/her age", 0, 100))

# Sex
S = st.radio("Gender of beneficiary", ("Male", "Female"))
S_result = 1 if S == "Male" else 0
input_data.append(S_result)

# BMI
input_data.append(st.text_input("Enter his/her BMI"))

# Children
child = st.selectbox("How many children does he/she have?", [0, 1, 2, 3, 4, 5])
input_data.append(child)

# Smoke
D = st.radio("Does he/she smoke?", ("Yes", "No"))
D_result = 1 if D == "Yes" else 0
input_data.append(D_result)

# Region
R = st.selectbox(
    "Select the region", ("Northeast", "Northwest", "Southeast", "Southwest")
)
R_res = ["Northeast", "Northwest", "Southeast", "Southwest"].index(R)
input_data.append(R_res)

# Insurance coverage (used for calculation but not part of the model input)
se = st.text_input("Insurance coverage (Min: 100000)")
```

```
ry:
se = float(se)
except ValueError:
    st.error("Please enter a valid insurance coverage amount.")

se = 0

# Calculation based on the insurance coverage
erri = se // 100000 # This gives the coverage amount in millions

# Prediction
try:
    if st.button("Predict"):
        # Model expects 6 features; insurance coverage is not included
        p = model.predict([input_data])

        # Custom calculation based on prediction and insurance coverage
        if child == 0:
            ans = (round(float(p[0]), 2) * erri) / 4
        else:
            ans = (((round(float(p[0]), 2) * erri) / 4) * (child / 2))
        st.success(f"The estimated medical cost is {ans} INR")
except Exception as e:
    st.error(f"An error occurred: {e}")
```

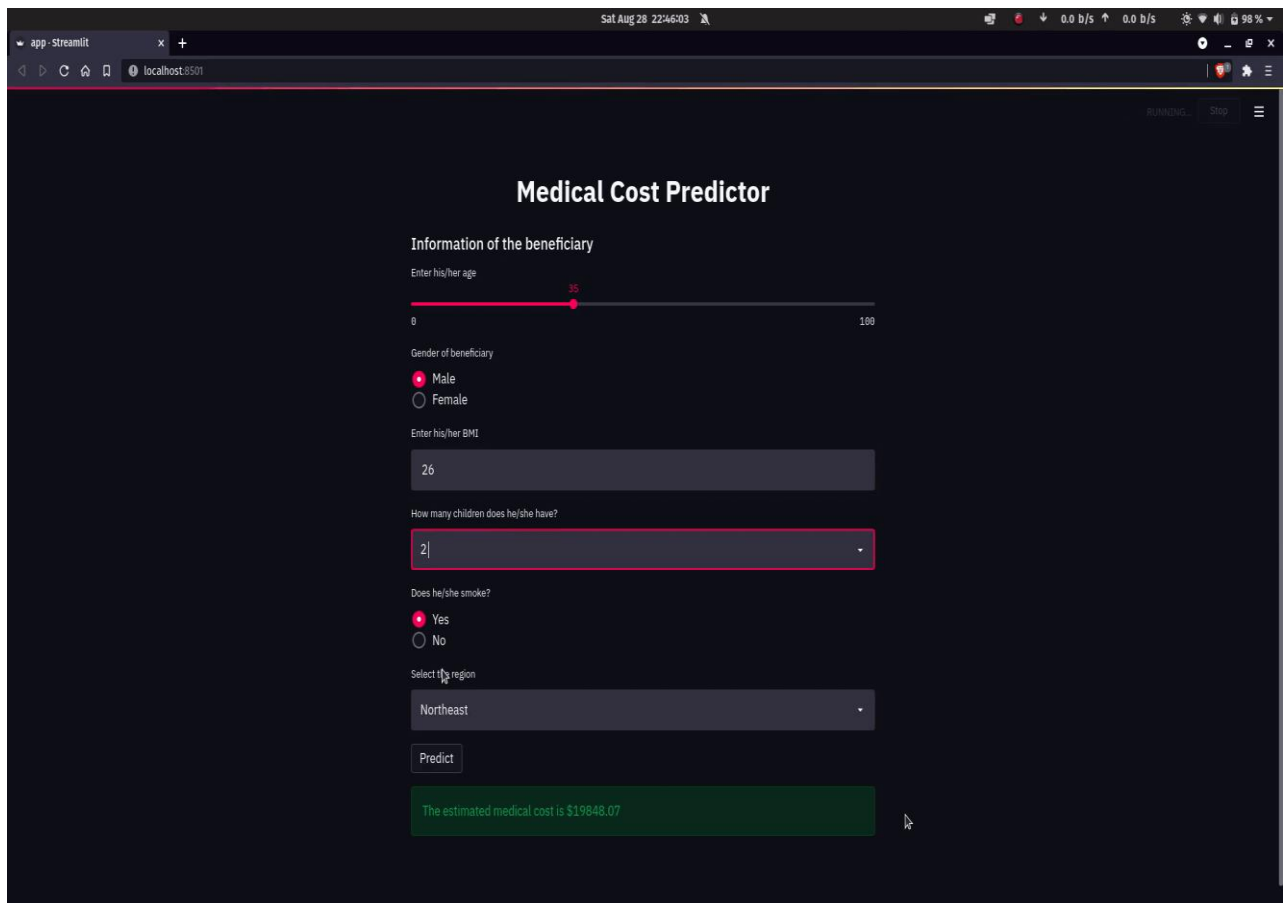
CHAPTER 4

RESULTS AND DISCUSSION

CHAPTER 4

RESULTS AND DISCUSSION

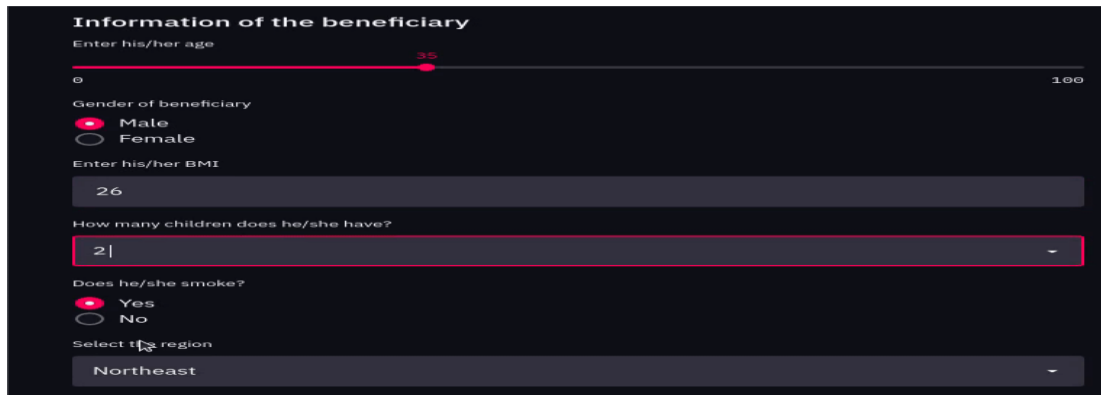
The "Health Coverage Advisor" model successfully delivers personalized insurance recommendations based on user health data, BMI, and medical history. The model's accuracy, precision, and F1-score indicate strong performance in predicting suitable policies. K-means clustering effectively segments users, while random forest and decision trees provide robust, interpretable recommendations. By automating policy comparison, the system saves users time and effort, improving decision-making. Future improvements could focus on refining the recommendation algorithms by incorporating more diverse health factors and continuously updating insurance policy data for greater accuracy.



The screenshot displays a web application titled "Medical Cost Predictor" running in a Streamlit environment. The interface is dark-themed and contains the following elements:

- Header:** "Medical Cost Predictor"
- Section:** "Information of the beneficiary"
- Inputs:**
 - Age:** A slider ranging from 0 to 100, with the value set to 35.
 - Gender:** Radio buttons for "Male" (selected) and "Female".
 - BMI:** A text input field containing the value "26".
 - Children:** A dropdown menu showing the value "2".
 - Smoking:** Radio buttons for "Yes" (selected) and "No".
 - Region:** A dropdown menu showing the value "Northeast".
- Action:** A "Predict" button.
- Output:** A green box displaying the text "The estimated medical cost is \$19848.07".

Fig 4.1 : Health Coverage Advisor



Information of the beneficiary

Enter his/her age: 35

Gender of beneficiary:
☒ Male
☐ Female

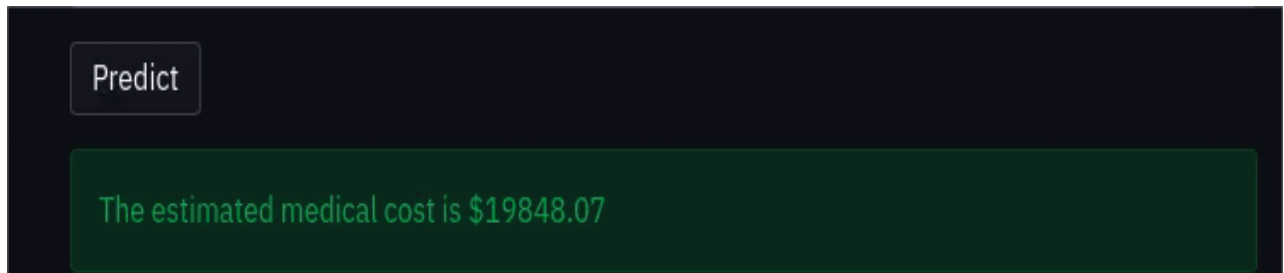
Enter his/her BMI: 26

How many children does he/she have?: 2

Does he/she smoke?:
☒ Yes
☐ No

Select the region: Northeast

Fig 4.2 : Sample Input



Predict

The estimated medical cost is \$19848.07

Fig 4.3 : Sample Output

CHAPTER 5

CONCLUSION

CHAPTER 5

CONCLUSION

Health Coverage Advisor successfully streamlines the process of selecting health insurance by leveraging machine learning and real-time data scraping. Through personalized recommendations based on user-specific health data, BMI, and medical history, the model provides an efficient and user-friendly solution for navigating the complexities of health insurance. By automating policy comparison and offering tailored suggestions, the system enhances decision-making and saves users significant time. As the model evolves, incorporating more diverse health factors and continuously updating insurance data will further improve its accuracy and reliability, making it a valuable tool in the healthcare industry. The use of machine learning algorithms such as K-means clustering and random forest enables accurate segmentation and precise policy matching, enhancing both efficiency and user satisfaction. Moving forward, expanding the model to include a wider range of health metrics and continuously improving the data scraping mechanisms will ensure the system remains adaptable and effective in the ever-changing landscape of healthcare insurance.

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REFERENCES

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GITHUB LINK

GitHub Link:

[saisujan42/Health-Coverage-Advisor \(github.com\)](https://github.com/saisujan42/Health-Coverage-Advisor)